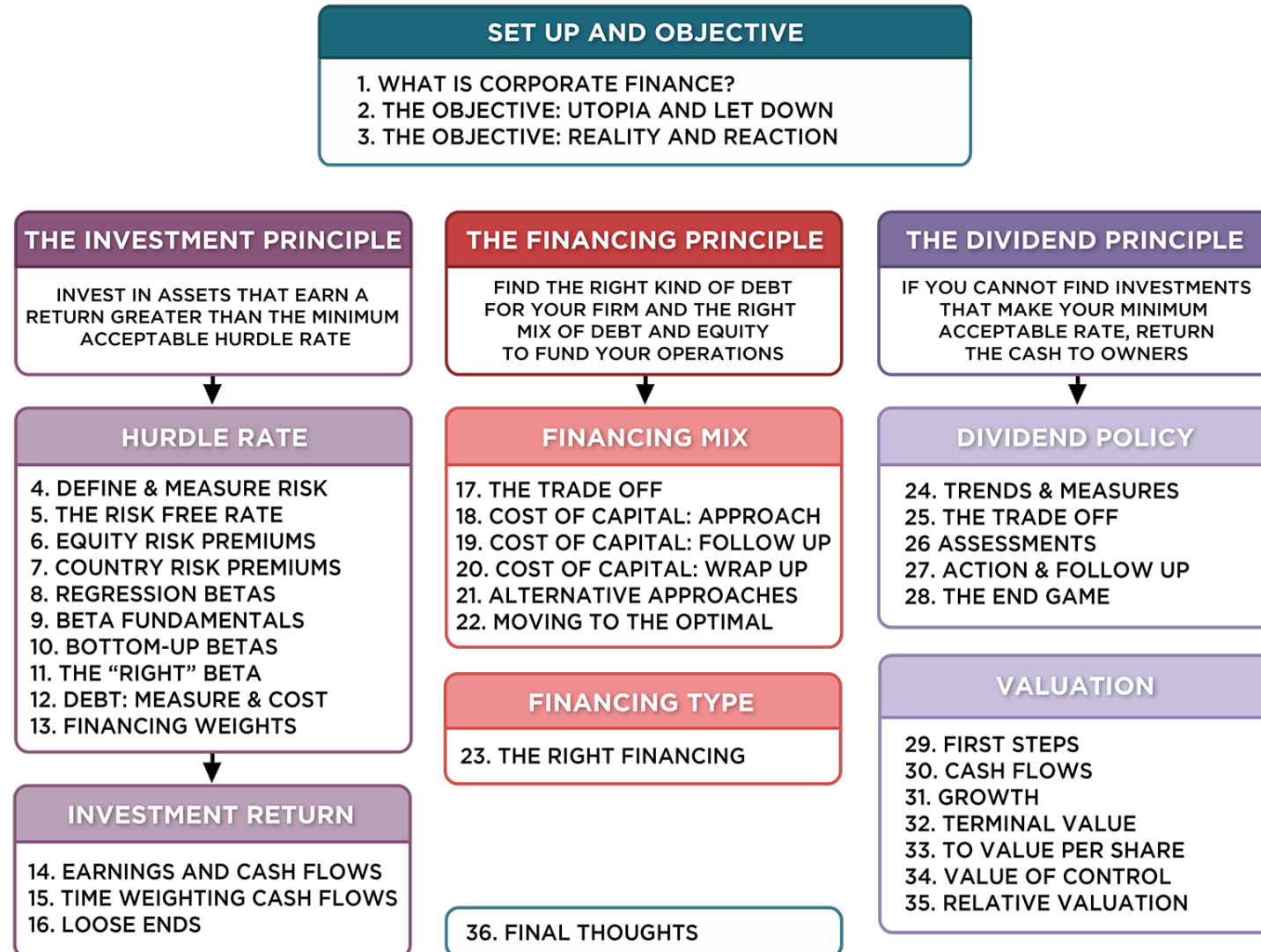


SESSION 18: THE COST OF CAPITAL APPROACH

It is better to have a lower hurdle rate than a higher one.

SESSION 18: THE COST OF CAPITAL APPROACH



THE COST OF CAPITAL APPROACH

- Value of a Firm = Present Value of Cash Flows to the Firm, discounted back at the cost of capital
- If the cash flows to the firm are held constant, and the cost of capital is minimized, the value of the firm will be maximized.

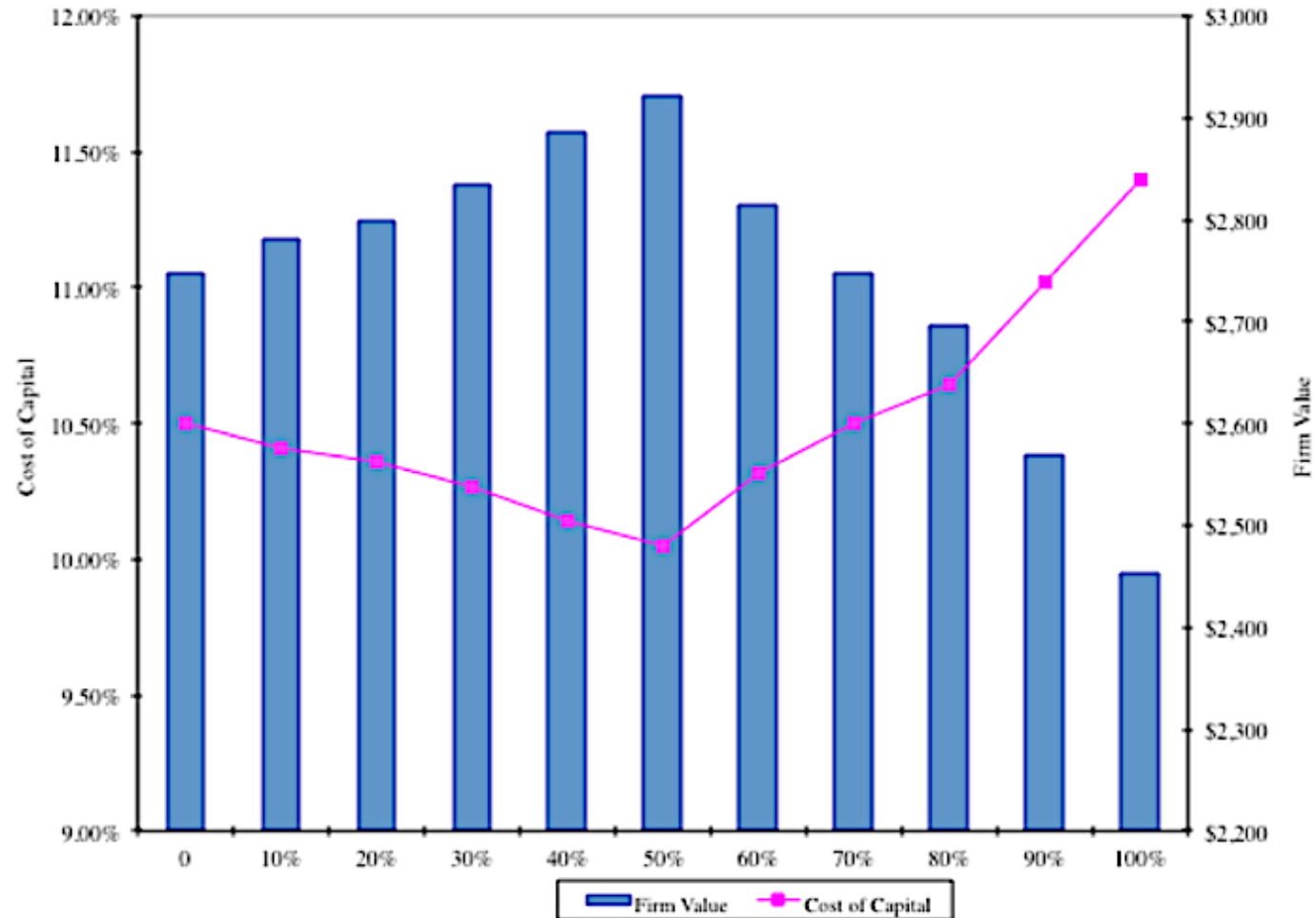
APPLYING COST OF CAPITAL APPROACH: THE TEXTBOOK EXAMPLE

D/(D+E)	Cost of Equity	After-tax Cost of Debt	Cost of Capital	Firm Value
0	10.50%	4.80%	10.50%	\$2,747
10%	11.00%	5.10%	10.41%	\$2,780
20%	11.60%	5.40%	10.36%	\$2,799
30%	12.30%	5.52%	10.27%	\$2,835
40%	13.10%	5.70%	10.14%	\$2,885
50%	14.50%	6.10%	10.30%	\$2,822
60%	15.00%	7.20%	10.32%	\$2,814
70%	16.10%	8.10%	10.50%	\$2,747
80%	17.20%	9.00%	10.64%	\$2,696
90%	18.40%	10.20%	11.02%	\$2,569
100%	19.70%	11.40%	11.40%	\$2,452

$$\frac{\text{Expected Cash flow to firm next year}}{(\text{Cost of capital} - g)} = \frac{200(1.03)}{(\text{Cost of capital} - g)}$$

THE U-SHAPED COST OF CAPITAL GRAPH

Figure 8.2: Cost of Capital and Firm Value



CURRENT COST OF CAPITAL: DISNEY

- The beta for Disney's stock in November 2013 was 1.0013. The T-Bond rate at that time was 2.75%. Using an estimated equity risk premium of 5.76%, we estimated the cost of equity for Disney to be 8.52%:

$$\text{Cost of Equity} = 2.75\% + 1.0013 (5.76\%) = 8.52\%$$

- Disney's bond rating in May 2009 was A, and based on this rating, the estimated pre-tax cost of debt for Disney is 3.75%. Using a marginal tax rate of 36.1, the after-tax cost of debt for Disney is 2.40%.

$$\text{After-Tax Cost of Debt} = 3.75\% (1 - 0.361) = 2.40\%$$

- The cost of capital was calculated using these costs and the weights based on market values of equity (121,878) and debt (15,961):

$$\text{Cost of capital} =$$

MECHANICS OF COST OF CAPITAL ESTIMATION

1. Estimate the Cost of Equity at different levels of debt:
 - Equity will become riskier → Beta will increase → Cost of Equity will increase
 - Estimation will use levered beta calculation
2. Estimate the Cost of Debt at different levels of debt:
 - Default risk will go up and bond ratings will go down as debt goes up → Cost of Debt will increase
 - To estimate bond ratings, we will use the interest coverage ratio ($\text{EBIT} / \text{Interest expense}$)
3. Estimate the Cost of Capital at different levels of debt
4. Calculate the effect on Firm Value and Stock Price

LAYING THE GROUNDWORK:

1. ESTIMATE THE UNLEVERED BETA FOR THE FIRM

- One approach is to use the regression beta (1.25) and then unlever, using the average debt to equity ratio (19.44%) during the period of the

$$\text{Unlevered beta} = 1.25 / [1 + (1 - 0.361)(0.1944)] = 1.1119$$

- Alternatively, we can back to the source and estimate it from the betas of the businesses.

<i>Business</i>	<i>Revenues</i>	<i>EV/Sales</i>	<i>Value of Business</i>	<i>Proportion of Disney</i>	<i>Unlevered beta</i>	<i>Value</i>	<i>Proportion</i>
Media Networks	\$20,356	3.27	\$66,580	49.27%	1.03	\$66,579.81	49.27%
Parks & Resorts	\$14,087	3.24	\$45,683	33.81%	0.70	\$45,682.80	33.81%
Studio Entertainment	\$5,979	3.05	\$18,234	13.49%	1.10	\$18,234.27	13.49%
Consumer Products	\$3,555	0.83	\$2,952	2.18%	0.68	\$2,951.50	2.18%
Interactive	\$1,064	1.58	\$1,684	1.25%	1.22	\$1,683.72	1.25%
Disney Operations	\$45,041		\$135,132	100.00%	0.9239	\$135,132.11	100.00%

2. GET DISNEY'S CURRENT FINANCIALS . . .

	Most recent fiscal year (2012-13)	Prior year
Revenues	\$45,041	\$42,278
EBITDA	\$10,642	\$10,850
Depreciation & Amortization	\$2,192	\$1,987
EBIT	\$9,450	\$8,863
Interest Expenses	\$349	\$564
EBITDA (adjusted for leases)	\$12,517	\$11,168
Depreciation (adjusted for leases)	\$ 2,485	\$2,239
EBIT (adjusted for leases)	\$10,032	\$8,929
Interest Expenses (adjusted for leases)	\$459	\$630

I. COST OF EQUITY

Debt to Capital Ratio	D/E Ratio	Levered Beta	Cost of Equity
0%	0.00%	0.9239	8.07%
10%	11.11%	0.9895	8.45%
20%	25.00%	1.0715	8.92%
30%	42.86%	1.1770	9.53%
40%	66.67%	1.3175	10.34%
50%	100.00%	1.5143	11.48%
60%	150.00%	1.8095	13.18%
70%	233.33%	2.3016	16.01%
80%	400.00%	3.2856	21.68%
90%	900.00%	6.2376	38.69%

Levered Beta = $0.9239 (1 + (1 - .361) (D/E))$

Cost of equity = $2.75\% + \text{Levered beta} * 5.76\%$

ESTIMATING COST OF DEBT

Start with the market value of the firm = $121,878 + \$15,961 = \$137,839$ million

D/(D+E)	0.00%	10.00%	Debt to capital
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D/E	0.00%	11.11%	D/E = $10/90 = .1111$
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\$ Debt	\$0	\$13,784	10% of \$137,839
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EBITDA	\$12,517	\$12,517	Same as 0% debt
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Depreciation	\$ 2,485	\$ 2,485	Same as 0% debt
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EBIT	\$10,032	\$10,032	Same as 0% debt
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Interest	\$0	\$434	Pre-tax cost of debt * \$ Debt
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Pre-tax Int. cov ∞	23.10	EBIT/ Interest Expenses
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Likely Rating	AAA	AAA From Ratings table
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Pre-tax cost of debt	3.15%	3.15%	Riskless Rate + Spread
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THE RATINGS TABLE

Large cap (>\$5 billion)	Ratings (S&P/Moody's)	Spread (11/13)	Interest Rate
>8.50	Aaa/AAA	0.40%	3.15%
6.5-8.5	As2/AAA	0.70%	3.45%
5.5-6.5	A1/A+	0.85%	3.60%
4.25-5.5	A2/A	1.00%	3.75%
3-4.25	A3/A-	1.30%	4.05%
2.5-3	Baa2/BBB	2.00%	4.75%
2.25-2.5	Ba1/BB+	3.00%	5.75%
2-2.25	Ba2/BB	4.00%	6.75%
1.75-2.25	B1/B+	5.50%	8.25%
1.5-1.75	B2/B	6.50%	9.25%
1.25-1.5	B3/B-	7.25%	10.00%
0.8-1.25	Caa/CCC	8.75%	11.50%
0.65-0.8	Ca2/CC	9.50%	12.25%
0.2-0.65	C2/C	10.50%	13.25%
<0.2	D2/D	12.00%	14.75%

T-Bond rate = 2.75%

A TEST: CAN YOU DO THE 30% LEVEL?

		<i>Iteration 1</i> <i>(Debt @AAA rate)</i>	<i>Iteration 2</i> <i>(Debt @AA rate)</i>
$D/(D + E)$	20.00%	30.00%	30.00%
D/E	25.00%	30/70=42.86%	
\$ Debt	\$27,568	\$41,352	
EBITDA	\$12,517	\$12,517	
Depreciation	\$2,485	\$2,485	
EBIT	\$10,032	\$10,032	
Interest expense	\$868	\$1,302	\$1,427
Interest coverage ratio	11.55	7.70	7.03
Likely rating	AAA	AA	AA
Pretax cost of debt	3.15%	3.45%	3.45%

BOND RATINGS, COST OF DEBT AND DEBT RATIOS

Debt Ratio	\$ Debt	Interest Expense	Interest Coverage Ratio	Bond Rating	Pre-tax cost of debt	Tax rate	After-tax cost of debt
0%	\$0	\$0	∞	Aaa/AAA	3.15%	36.10%	2.01%
10%	\$13,784	\$434	23.10	Aaa/AAA	3.15%	36.10%	2.01%
20%	\$27,568	\$868	11.55	Aaa/AAA	3.15%	36.10%	2.01%
30%	\$41,352	\$1,427	7.03	Aa2/AA	3.45%	36.10%	2.20%
40%	\$55,136	\$2,068	4.85	A2/A	3.75%	36.10%	2.40%
50%	\$68,919	\$6,892	1.46	B3/B-	10.00%	36.10%	6.39%
60%	\$82,703	\$9,511	1.05	Caa/CCC	11.50%	36.10%	7.35%
70%	\$96,487	\$11,096	0.90	Caa/CCC	11.50%	32.64%	7.75%
80%	\$110,271	\$13,508	0.74	Ca2/CC	12.25%	26.81%	8.97%
90%	\$124,055	\$16,437	0.61	C2/C	13.25%	22.03%	10.33%

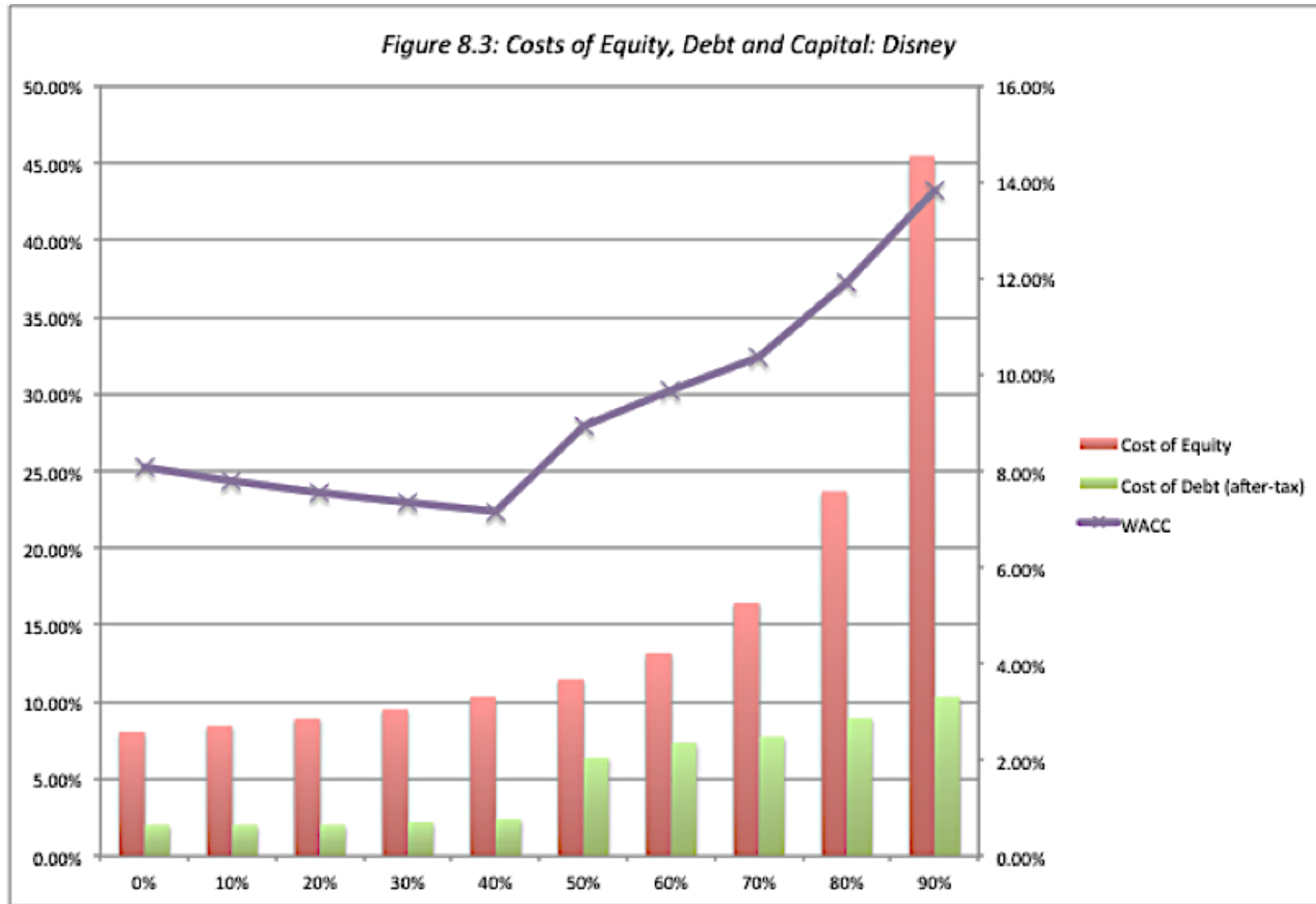
STATED VERSUS EFFECTIVE TAX RATES

- You need taxable income for interest to provide a tax savings. Note that the EBIT at Disney is \$10,032 million. As long as interest expenses are less than \$10,032 million, interest expenses remain fully tax-deductible and earn the 36.1% tax benefit. At a 60% debt ratio, the interest expenses are \$9,511 million and the tax benefit is therefore 36.1% of this amount.
- At a 70% debt ratio, however, the interest expenses balloon to \$11,096 million, which greater than the EBIT of \$10,032 million. We consider the tax benefit on the interest expenses up to this amount:
 - $\text{Maximum Tax Benefit} = \text{EBIT} \times \text{Marginal Tax Rate} = \$10,032 \text{ million} \times 0.361 + \$3,622 \text{ million}$
 - $\text{Adjusted Marginal Tax Rate} = \text{Maximum Tax Benefit} / \text{Interest Expenses} = \$3,622 / \$11,096 = 32.64\%$

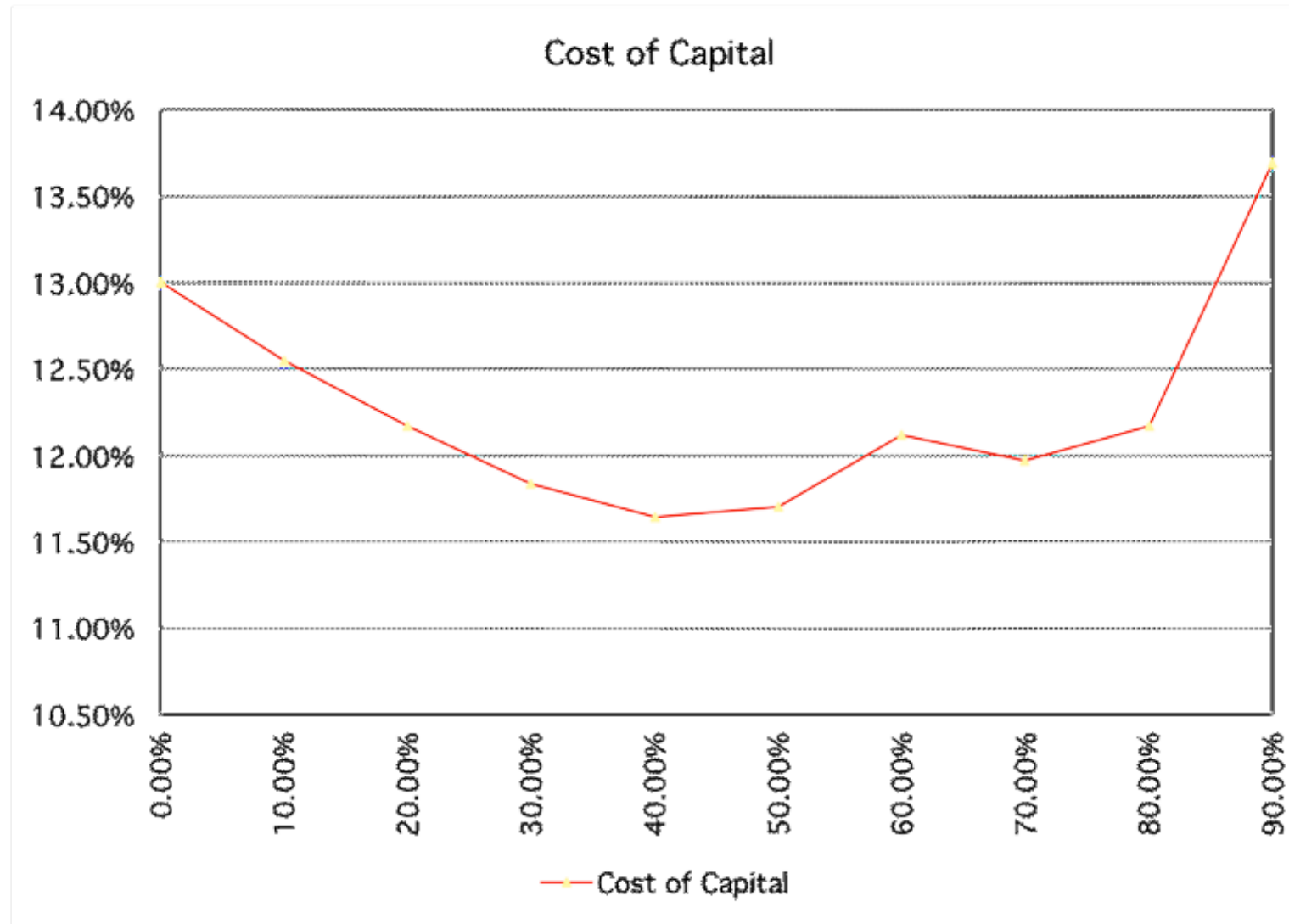
DISNEY'S COST OF CAPITAL SCHEDULE . . .

Debt Ratio	Beta	Cost of Equity	Cost of Debt (after-tax)	WACC
0%	0.9239	8.07%	2.01%	8.07%
10%	0.9895	8.45%	2.01%	7.81%
20%	1.0715	8.92%	2.01%	7.54%
30%	1.1770	9.53%	2.20%	7.33%
40%	1.3175	10.34%	2.40%	7.16%
50%	1.5143	11.48%	6.39%	8.93%
60%	1.8095	13.18%	7.35%	9.68%
70%	2.3762	16.44%	7.75%	10.35%
80%	3.6289	23.66%	8.97%	11.90%
90%	7.4074	45.43%	10.33%	13.84%

DISNEY: COST OF CAPITAL CHART



DISNEY: COST OF CAPITAL CHART – 1997



Task

Use the cost of capital approach to estimate the optimal financing mix for your company

FOURTH EDITION

Applied Corporate Finance

ASWATH DAMODARAN



WILEY

Optional:
Read
Chapter 8